

Introduction to Basic Mathematics

A Foundation Course for Confident Numeracy and Practical Problem Solving

Course Overview

Mathematics is the foundation of all technical and scientific thinking. Whether it is reading a flow-meter reading, checking a water-quality report, budgeting for supplies, or interpreting a graph in a board presentation, sound numerical skills make every task faster, more reliable, and more professional. This self-paced foundation course builds those core numeracy skills from the ground up — using plain language, fully worked examples, and realistic workplace scenarios throughout. No prior knowledge beyond basic counting is assumed.

Who Is This Course For?

- Mid-secondary scholars (ages 14–18) looking to strengthen their numeracy before examinations or further study.
- Water-sector technicians, operational staff, and administrative employees undertaking upskilling or re-skilling as part of workplace development.
- Adult learners returning to study who want a practical, jargon-free refresher in core mathematics.
- Anyone preparing for further technical, vocational, or academic training that assumes solid numerical foundations.

Course Purpose & Structure

This course takes learners on a carefully sequenced journey — from understanding numbers and basic arithmetic, through fractions, decimals and percentages, to reading and interpreting data for practical reporting. The Introduction to Basic Mathematics self-study course will assist participants with basic data analyses and report writing. It is delivered entirely online via the WBC Tutor LMS, so learners can study at their own pace, revisit any topic at any time, and track their own progress.

Every one of the 10 topics follows the same consistent, easy-to-follow structure:

- Learning Objectives & Key Concepts — know exactly what you'll be able to do by the end of the topic.
- Definition & Explanation — plain-language teaching with clearly labelled examples and figures.
- Key Mathematical Rules — the essential rules, numbered for quick reference.
- Worked Examples — step-by-step solutions, including practical workplace applications.
- Self-Study Practice — problems to attempt independently, with full worked solutions provided.
- Tips for Learners — practical advice to help you avoid the most common mistakes.
- Diagnostic Quiz — an auto-marked quiz covering every rule taught, so you can check your understanding before moving on.

A companion Quick Reference Sheet summarises every key rule and formula on a single page — ideal for ongoing use once the course is complete. Altogether, the course represents approximately 32 to 40 hours of self-paced study. However, the total time allocated for online access to complete the self-study course on the WBC LMS is two (2) months.

What You'll Learn — Topics & Objectives

No.	Topic	What You'll Learn
1	Exponent — Power Notation	Interpret and apply power notation (base and exponent); evaluate expressions involving positive, negative and zero exponents; apply exponent rules to simplify numerical expressions.
2	Scientific Notation	Express very large and very small numbers in scientific notation; convert between standard and scientific notation; perform multiplication and division using scientific notation in technical contexts.
3	Dimensional Analysis	Use dimensional analysis (factor-label method) to convert between units of measurement; verify the consistency of units in equations; apply unit conversions to practical and technical problems.
4	Rounding	Round numbers to a specified number of decimal places or significant figures; apply appropriate rounding conventions in measurement and reporting; recognise the impact of rounding on accuracy.
5	Estimating	Apply estimation strategies to predict reasonable answers before calculating; use front-end estimation, rounding, and compatible numbers; evaluate whether a calculated result is realistic in context.
6	Universal Rules for Solving Equations	State and apply the fundamental rules for manipulating equations (addition, subtraction, multiplication and division of both sides); maintain equation balance and verify solutions by substitution.
7	Solving for the Unknown Value	Solve one-step and multi-step equations to find an unknown variable; rearrange formulas to make a specific variable the subject; apply equation-solving techniques to technical and workplace problems.

No.	Topic	What You'll Learn
8	Mensuration	Calculate perimeter, area and volume of common two- and three-dimensional shapes; select and apply the correct formula for each shape; interpret mensuration results in practical measurement contexts.
9	Descriptive Statistics	Calculate and interpret measures of central tendency (mean, median, mode) and spread (range); organise data in tables and frequency distributions; present and communicate statistical findings for report writing and basic data analyses.
10	Functions and Graphs	Understand the concept of a function and function notation; plot and interpret graphs for basic straight-line (linear), quadratic and exponential functions; read values from graphs and identify trends; apply graphical interpretation to real-world data sets.

Course Policies & Practical Information

1. Engagement with Facilitator

The learner / participant will be able to post questions and/or comments on the activities / tasks. The developer of this self-study course will regularly respond to the questions and comments. The developer may also arrange an online meeting to respond to the questions and/or comments. The date and time for such an online meeting, specific to a course, will be posted on the WBC LMS. All online meetings / forums, etc may / will be recorded and posted on the WBC LMS for the benefit of all learners / participants registered for the course.

2. Flexibility / Accessibility of Course Material

The self-study courses are available 24/7 and the learner / participant can start at any time once registered!

3. Course Structure

The content for this self-study course is sub-divided based on specific topics. Each topic is concluded with a quiz. Learners are expected to achieve 100% for each quiz. The course content for a subsequent topic can only be accessed if the learner achieves a 100% pass for a preceding quiz. The accompanying text and compulsory quizzes will guide learners through various topics including tasks / activities) to be completed by the learner.

4. Refunds

Cancelling and/or not completing the course does not qualify the learner / participant to be eligible for a refund. Refunds will only be considered if there are compelling reasons (for cancellation / termination) beyond the control of the learner / participant.

5. In Addition

- ❖ The Foundation Courses are not related to any qualification but may serve to support specific content of a related qualification.
- ❖ The Foundation Courses are not associated with specific NQF levels.
- ❖ This Foundation Courses are not accredited by the Engineering Council of South Africa (ECSA) and/ or the South African Council for Natural Scientific Professions (SACNASP).

6. Course Fees

The introductory fee for the online **Introduction to Basic Mathematics** self-study course **for employed staff is R 500**.

The **significantly reduced introductory fee** for the online **Introduction to Basic Mathematics** self-study course **for registered (current registrations) full-time and part-time scholars** at any education institution is **R 100**. Scholars will be required to provide a copy of their annual registration form (current year) issued by the educational institution. Please submit your documentation to:

students@waterbusinesscollege.co.za.

Ready to build your mathematics confidence?

Enrol today through the WBC Tutor LMS and start learning at your own pace.